

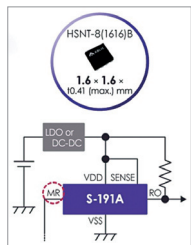
accuracy and fast measuring speeds, alongside a broad measuring range. It stands out as one of the smallest distance sensors in the radar market, making it suitable for tight spaces and short measuring distances. The sensor is suited for the automation of outdoor applications in agriculture, mining, transport and logistics as well as for factory automation in the manufacturing industry. The sensor has an accuracy of $\pm 2\text{mm}$ and measures at 100Hz, fitting for various applications. Its measuring range is from 0.2 to 40 metres.

OndoSense

<https://ondosense.com>

Automotive window voltage detectors with rapid response

ABLIC has launched the S-191A series of automotive window voltage detectors. The window voltage detector monitors undervoltage and overvoltage.



It has a response time of 10 μs from anomaly detection to output and includes a manual reset function for diagnosing faults. These features enhance functional

safety in automotive applications. The series boasts a rapid response time of 10 μs (typical) from undervoltage or overvoltage detection to output, placing it among the fastest in the industry. Equipped with a manual reset function, the series can diagnose non-safe IC faults. The detector series is housed in an ultra-compact package (1.6mm x 1.6mm x 0.41mm), significantly reducing its footprint by 70% compared to ABLIC's previous products.

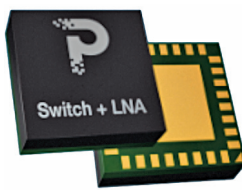
ABLIC

<https://www.ablic.com>

Dual-channel switch module for wireless applications

The pSemi Corporation has introduced a multi-chip module with dual-channel

switch, PE53230, designed for the 3.3–3.8GHz range. It includes two switches and two low noise amplifiers



(LNAs) catering to both high- and low-band frequencies. The module sets a new indus-

try standard with a noise figure of less than 1dB, delivering outstanding receiver sensitivity and performance while handling an average input power of 20W, eliminating the need for external circuitry. The module is designed for various wireless infrastructure applications, including TDD-LTE macro and micro cells.

pSemi Corporation

<https://www.psemi.com>

World's first free-shaped organic photovoltaic module

Dracula Technologies has launched LAYER OPV modules with a 25% performance improvement over the previous version. This advancement enables customers in IoT, wearables,



and ultra-low-power applications to generate 25% more power in the

same space or achieve the same power with 25% less surface area. Reduced surface area results in cost savings and enhances environmental sustainability, leading to discreet IoT designs, reduced maintenance expenses, lower labour costs, and reduced environmental footprints. The company claims that it stands as the world's first free-shaped OPV module, created through inkjet printing technology, capable of efficiently harnessing energy from ambient light. The module can take on various shapes, adapt to small formats, and use different materials to meet the needs of a range of applications.

Dracula Technologies

<https://dracula-technologies.com>

First IBM quantum processor with over 1000 qubits

IBM has launched Condor, a quantum processor with 1121 qubits. The qubits are arranged in a unique honeycomb pattern. These characteristics allow qubits



to exist in multiple states simultaneously, enabling computa-

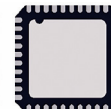
tions beyond the capabilities of classical computers. The company acknowledges the importance of error-correction techniques for practical quantum computing in the newly launched processor. The Heron chip, as part of this new processor, features 133 qubits with a low error rate, three times lower than its predecessor. Also, an alternative error-correction scheme, known as quantum low-density parity check, has the potential to reduce the required number of physical qubits for error correction significantly.

IBM

<https://www.ibm.com/in-en>

Integrated GNSS chip series with enhanced performance

Allystar has launched HD8140 series built on the state-of-the-art CYNOSURE IV SoC architecture. It combines two microprocessors, multi-band multi-



system GNSS RF and baseband components, two memory

units, an intelligent power management unit, and peripheral interfaces into a single chip. The design offers integration, robust functionality, low power consumption, and a compact size. The series finds applications in a wide range of fields, including mobile phones, wearables, unmanned aerial vehicles, vehicle navigation and management, autonomous driving, marine navigation, GIS data collection, and engineering surveys.

Allystar

<https://www.allystar.com>